Energy Management System for Parallel Mild Hybrid Vehicle Using Rule-Based Strategy and Dynamic Programming

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Abstract*—As the demand for sustainable and efficient transportation escalates, the role of energy management systems (EMS) in Hybrid and Electric Vehicles (HEVs) becomes paramount. This research pivots on the Mercedes-Benz A 170 CDI, a Parallel Mild Hybrid Electric Vehicle, focusing on enhancing its fuel efficiency via innovative EMS strategies. The first part of the study employs a Rule-Based Strategy (RBS), resulting in fuel consumption reduction with an efficiency of $27.32\,\%$ and $30.01\,\%$ for the New European Drive Cycle (NEDC) and US Federal Test Procedure (FTP-75) respectively when compared to a conventional vehicle. Further to this, a Dynamic Programming (DP) approach is introduced to optimize the EMS, delivering fuel consumption reduction with efficiency improvements of $33.32\,\%$ and $35.87\,\%$ for the NEDC and FTP-75 cycles respectively, when compared to a conventional vehicle. Crucially, the DP approach exhibits approximately a $6\,\%$ efficiency increase compared to the RBS. The comparisons are drawn based on equivalent fuel consumption, which provides a consistent benchmark for assessing EMS performance. Beyond the reduction in fuel consumption, both strategies are proven to sustain the battery life after the completion of driving cycles, signifying their practical applicability. The study utilizes MATLAB/Simulink as a platform for simulation and modeling, demonstrating its versatility in handling complex system interactions. Overall, this research provides valuable insights into the potential of EMS strategies to boost HEV efficiency, thereby contributing to the overarching goal of sustainable mobility.

***Index Terms*—Hybrid electric vehicles, rule-based strategy, dynamic programming, NEDC, FTP-75.**

I. Introduction

IT is projected that there will be a significant rise in the global vehicle count, with a projected rise from 700 million to 2.5 billion in the forthcoming years [1]. The underlying cause of this phenomenon can be attributed to the steady growth of the global population. If all vehicles were to rely solely on internal combustion engines, it would result in a significant depletion of oil and fossil fuel resources [1]. Furthermore, it is anticipated that global warming will continue to escalate and surpass the established threshold, resulting in a perpetual grayish appearance of the sky [1]. After considering all of these factors, it becomes evident that the most viable course of action is to prioritize the adoption of environmentally friendly energy sources. One potential solution to address these challenges is the implementation of electric vehicles, which offer both efficiency and reliability. Electric vehicles (EVs) are a type of automobile that operate using electrical propulsion. They can be categorized into three main types: Battery Electric Vehicles (BEVs), Hybrid Electric Vehicles (HEVs), and Fuel-Cell Electric Vehicles (FCEVs). The primary emphasis of this study is on hybrid electric vehicles [1].

Hybrid Electric Vehicles (HEVs) are comprised of an Internal Combustion Engine (ICE) in conjunction with a minimum of one electric motor [2]. The battery serves the purpose of storing electrical energy, which can subsequently be utilized for propelling vehicles as required [2]. Hybrid electric vehicles (HEVs) have the capability to function autonomously using an internal combustion (IC) engine, autonomously using an electric motor, or in a parallel combination of both [2]. One notable benefit of this technology is the potential for energy recovery through battery recharging [2]. The aforementioned attribute is commonly referred to as ”Regeneration,” which consequently leads to an extended driving range and a notable decrease in fuel consumption [2].

A previous study was undertaken to comprehensively examine various Energy Management Systems (EMSs) and their potential consequences [3]. The paper presented three distinct methodologies for energy management, each with the objective of enhancing overall performance and increasing energy efficiency [3]. A comprehensive examination was conducted to thoroughly analyze the current state of electric and hybrid vehicles worldwide [1]. The authors also took into account the most recent developments in Electric Vehicles and Hybrid Electric Vehicles (HEVs), with a particular focus on relevant engineering principles and technologies [1]. A separate study was conducted to showcase the implementation of a rule-based strategy. The objective of this study was to create a strategy that is both highly efficient and comparable in accuracy to existing methods, while significantly reducing the time required for computation [4]. The strategy employed by the researchers focused on attaining computational efficiency that was 3000 times faster, while simultaneously ensuring a tolerance level of 1 percent. This approach effectively optimized the drive train of the vehicle [10]. Furthermore, the study focused on examining the notion and classification of Hybrid Electric Vehicles (HEVs), emphasizing the significance of maintaining battery charge and implementing efficient battery management systems for HEVs [5]. A further extensive analysis of Dynamic Programming was conducted, which included a thorough investigation of the implementation procedure [3].

This study focuses on the Hybrid Electric Vehicle (HEV) utilizing the P2 architecture, which involves the placement of an electric motor between the clutch and gearbox. The configuration of the vehicle is summarized below.

- Vehicle Model - Mercedes-Benz A 170 CDI (W168)
- Engine Type - Diesel Engine (66 KW/ 180 Nm nominal, 60 KW/ 187 Nm measured)
- Electric Motor - 12 KW / 60 Nm
- Battery - Li-Ion, 16.38 KW / 0.468 KWh / 48 V

The objective of this study is to integrate an Energy Management System (EMS) into a MATLAB/Simulink model, with the aim of optimizing the vehicle's performance by operating in the most efficient manner possible. Several significant energy management strategies include the Rule-based strategy (RBS), Equivalent consumption minimization strategy (ECMS), Pontryagin's Minimum Principle (PMP), Dynamic Programming (DP), Stochastic Dynamic Programming (SDP), and Model Predictive Control (MPC) [2].

The Rule-based strategy (RBS) offers a straightforward and insightful method for achieving optimal power allocation between the battery and the engine across diverse driving scenarios. Furthermore, this technology enables the achievement of the desired optimal State of Charge (SOC) and superior fuel efficiency, along with various other design objectives. The Dynamic Programming (DP) approach is employed to achieve comprehensive optimal values and address the non-linearity, as RBS is found to be inadequate in delivering optimal results [3].

The construction of the paper is as follows: Section II emphasizes the classification of HEVs, the driving cycles considered in this paper, and the introduction to the QSS toolbox. The problem statement is defined in Section III. A detailed overview of operation modes is specified in section IV. Section V consists of detailed information about energy management strategies utilized in this study. Results and conclusions are presented in Section VI and Section VII respectively. The paper concludes with future scope in Section VIII

II. CLASSIFICATION OF HYBRID ELECTRIC VEHICLES, DRIVING CYCLES AND QSS TOOLBOX

A. Classification of Electric vehicles

1) Series Hybrid Electric Vehicles: According to the given configuration, the vehicle can only receive propulsion power from a single power source [5]. Primarily, the internal combustion engine functions as a primary means of propulsion [5]. In addition, the internal combustion engine (IC engine) is responsible for recharging the battery through an electric generator. This generator supplies power to both the battery and the propelling motor as required [5].

2) Parallel Hybrid Electric Vehicles: In the context of parallel hybrid electric vehicles (HEVs), it is possible for multiple power sources to contribute to the propulsion power that drives the vehicle's wheels [5]. In contrast to the Series HEV, it is important to note that there does not exist direct mechanical coupling between the internal combustion engine (IC engine) and the electric motor [5]. Consequently, both components are interconnected in a parallel configuration by means of a mechanical coupling [5]. The mechanical coupling in question serves to integrate the power derived from the power sources, as indicated by reference [5]. Given the potential to contribute to overall power, it has been observed that the power demand of an electric motor is comparatively lower when operating in a hybrid electric drive system, as opposed to a purely electric drive system [5].

3) Combined Hybrid Electric Vehicles: By combining two separate configurations, the benefits of both series and parallel hybrid electric vehicles (HEVs) can be obtained [5]. According to the referenced study, the series of hybrid electric vehicles (HEVs) demonstrate a high level of operational efficiency when operating at lower speeds. Conversely, parallel HEVs exhibit efficient performance at higher speeds. The complexity of the architecture increases as a result of the inclusion of supplementary mechanical links in comparison to series hybrid electric vehicles (HEVs), as well as the incorporation of generators in contrast to parallel HEVs [5].

B. Driving Cycles Used

1) New European Drive Cycle (NEDC): The New European Drive Cycle (NEDC) is a standardized test procedure used to measure the fuel consumption and emissions of vehicles in Europe. The driving cycle described in this study is primarily designed for synthetic driving profiles of light-duty vehicles, but it can also be applied to electric vehicles and hybrid electric vehicles (HEVs) [1]. In this cycle, vehicle auxiliaries, such as air conditioning, are not considered [2]. According to the cited source [2], the overall distance covered by this cycle is 11 kilometers, with a corresponding duration of 1220 seconds. According to the cited source, the vehicle has the capability to reach a maximum velocity of 120 km/h, while maintaining an average speed of 33.6 km/h. [2]

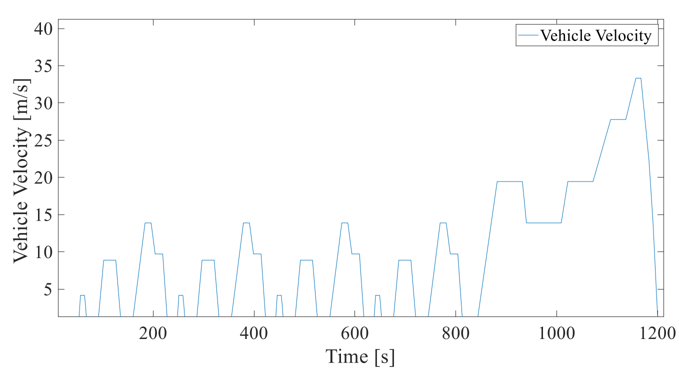

Fig. 1. NEDC [2]

2) US Federal Test Procedure-75 (FTP-75): The US Federal Test Procedure-75 (FTP-75) is a driving cycle that was developed in 2008 by integrating different driving cycles, including the highway driving cycle (HWFET), aggressive driving cycle (US06), and an optional air conditioning test (SC). This driving cycle serves as the primary source of measurement data. [2] The driving cycle in question has a distance of 17.77 kilometers and a duration of 1877 seconds [2]. This cycle has the capability to achieve a maximum speed of 91.2 km/h and an average speed of 34.1 km/h. An additional parameter known as ”hot Soak” can be incorporated within the time frame of 540-660 seconds during the cycle, as stated in reference [2].

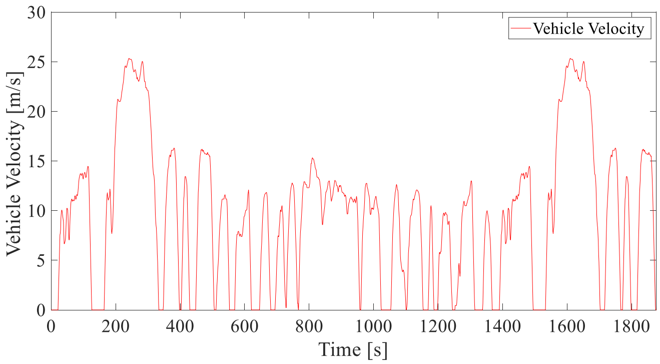

Fig. 2. FTP-75 [2]

C. QSS Toolbox

QSS Toolbox is a software tool that is utilized for various purposes in the field of quantitative systems safety. The acronym QSS refers to Quasi-static Modelling and Simulation. The aforementioned software package is a MATLAB/Simulink toolbox designed specifically for the purpose of modeling and simulating Hybrid Electric Powertrains [2]. The toolbox has the capability to be utilized with a range of driving cycles and control strategies in order to compute the most efficient fuel consumption [2]. The toolbox is founded upon the theoretical framework of reverse cause-and-effect interactions within dynamic systems [2]. The Toolbox presented in this study demonstrates a novel methodology for calculating acceleration and estimating forces by utilizing given speeds, as opposed to the traditional method of estimating speeds from provided forces [1]. The QSS toolbox used for the energy management study is shown in Fig. 3.

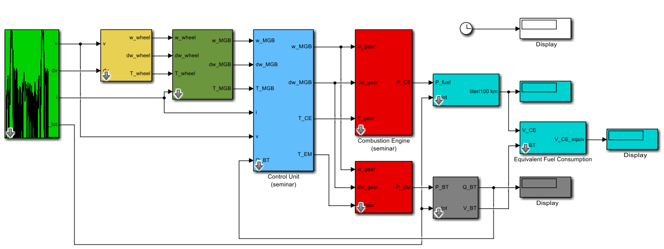

Fig. 3. QSS Toolbox

III. PROBLEM STATEMENT

Hybrid Electric Vehicles (HEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) are equipped with an Electric Power Source, which comprises a storage device, power electronics, and an electric motor, alongside a Combustion Engine. The primary objective of incorporating an electric motor into a combustion engine is to mitigate the consumption of fuel. The development and deployment of the Energy Management System should be executed in a manner that effectively reduces overall fuel consumption to a specific degree. In addition to the decrease in fuel consumption, it is imperative that the battery's State of Charge (SOC) remains consistent with its initial value at the conclusion of the cycle. The examination of different Drive Modes will also contribute to the fulfillment of requirements. A comparative analysis is conducted between the Rule-Based Strategy and Dynamic Programming, utilizing the results obtained from their implementation in the MATLAB/Simulink Model.

IV. OPERATION MODES

The objective of this study is to investigate the energy management strategies employed in the context of a Parallel Hybrid Vehicle. In parallel hybrid vehicles, the mechanical coupling of the two energy sources, namely the electric motor and combustion engine, is achieved through the use of a Torque Coupler. The calculation of gearbox torques is thus provided in (1).

$$T_{\text{MGB}} = T_{\text{EM}} + T_{\text{CE}} \tag{1}$$

where T_{CE} is the torque provided by combustion engine, and T_{EM} is the torque provided by electric motor, T_{MGB} is the flywheel torque. The ratio of torque provided by an electric motor to the flywheel torque is known as the Torque split ratio. Mathematically it is shown in (2).

$$u = T_{\text{EM}}/T_{\text{MGB}} \tag{2}$$

The determination of the operation mode of a vehicle is significantly influenced by the torque split ratio. The operations that were examined in this study are as follows.

A. Regeneration Mode

In conventional automobiles, the process of braking involves the conversion of kinetic energy into heat energy through the utilization of friction. The process of regeneration involves utilizing the motor in generator mode during braking to capture and store the kinetic energy in a battery for subsequent use [6]. Fig. 4 [2] presents the block diagram illustrating the energy flow in the regeneration mode. The condition for regeneration mode is $T_{\text{MGB}} < 0$. The torque Split Ratio (u) is found using the formula mentioned in (3).

$$u = \min\left(\frac{-T_{\text{EM}}, \max(\omega_{\text{EM}}) + |\theta_{\text{EM}} d\omega| + \varepsilon}{T_{\text{MGB}}}, 1\right) \tag{3}$$

B. Load Point Shifting

The efficiency of the engine, denoted as η_{CE}, is heavily influenced by the load points ω_{CE} and T_{CE}, where ω_{CE} represents the angular velocity of the combustion engine. In order to enhance the efficiency of the engine, the load points are adjusted by functioning the electric motor under

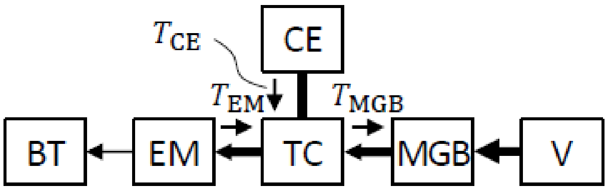

Fig. 4. Energy flow in regeneration mode [2]

specific desired conditions. In the motor mode, the electric motor provides power to the manual gearbox, resulting in a reduction in the load point. However, this also results in the depletion of the battery. In generator mode, the electric motor functions as a power generator, facilitating the charging of the battery and thereby enabling an increase in the load point. This can be achieved by ensuring that the improvement in efficiency consistently surpasses the losses incurred during the conversion process. Enhancing efficiency can be achieved by employing the electric motor in motor mode during low-load conditions and as a generator during medium-load conditions. The torque coupler is subject to the constraint that the sum of the torque from the engine (T_{CE}) and the torque from the motor (T_{EM}) is equal to the torque on the motor gearbox (T_{MGB}). Additionally, the engine torque (T_{CE}) must be greater than or equal to zero and less than or equal to the maximum engine torque ($T_{\mathrm{CE,\ max}}$) at a given angular velocity (ω_{CE}). Similarly, the motor torque (T_{EM}) must be less than the maximum motor torque ($T_{\mathrm{EM,\ max}}$) at a given angular velocity (ω_{EM}). These constraints also apply to the battery. Fig. 5 [2] depicts the energy flow in load point shifting mode.

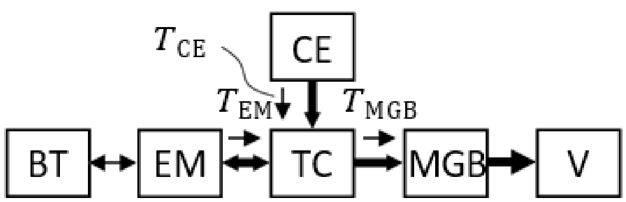

Fig. 5. Energy flow in Load Point Shifting mode [2]

1) Motor mode: The electric motor provides energy to the manually operated gearbox, resulting in a reduction in the load point. However, this process also results in the depletion of the battery. The condition for load point shifting under motor mode is ($T_{\mathrm{MGB}} \geq T_{\mathrm{MGB\_th}}$) && ($Q_{\mathrm{BT}} > Q_{\mathrm{BT\_max}}$). The torque Split Ratio (u) under motor mode is found using the formula mentioned in (4).

$$u = \min\left(\frac{-T_{\mathrm{EM}}, \max(\omega_{\mathrm{EM}}) - |\theta_{\mathrm{EM}} d\omega| + \varepsilon}{T_{\mathrm{MGB}}}, u_{\mathrm{LPS\_max}}\right) \quad (4)$$

2) Generator mode: The electric motor functions as a power generator, facilitating the charging of the battery. Consequently, the load point can be elevated. The condition for load point shifting under generator mode is($T_{\mathrm{MGB}} > T$) && ($T_{\mathrm{MGB}} < T_{\mathrm{MGB\_th}}$). The torque Split Ratio (u) under the generator mode is found using the formula in (5).

$$u = \max\left(\frac{-T_{\mathrm{EM}}, \max(\omega_{\mathrm{EM}}) + |\theta_{\mathrm{EM}} d\omega| + \varepsilon}{T_{\mathrm{MGB}}}, u_{\mathrm{LPS\_min}}\right) \quad (5)$$

C. Start-Stop

All of the operation modes mentioned earlier meet the necessary criteria while the vehicle is in motion. However, if the vehicle is idle, such as at traffic signals, the engine remains running at an idle state, resulting in energy loss. This occurs because the combustion engine continues to burn fuel to sustain its operation. In order to address this issue, the implementation of a start-stop system is employed, which halts the combustion engine when the vehicle is in an idle state, thereby conserving fuel. The system operates in conjunction with either an electric drive ($T_{\mathrm{MGB}} > 0$) or regeneration ($T_{\mathrm{MGB}} < 0$) prior to the vehicle reaching an idling state.

D. Electric Driving

The efficiency of a combustion engine significantly decreases when operating at low load points, such as reduced speeds with frequent stops. In order to address the issue of reduced efficiency, one potential solution is the integration of electric drive technology at lower operational points. Additionally, the implementation of charging mechanisms during medium load points can further enhance overall efficiency. The energy flow during electric driving mode is shown in Fig. 6 [2]. Additionally, the implementation of electric propulsion systems results in reduced emissions and decreased levels of noise pollution emitted by the vehicle.The condition for electric driving mode is ($T_{\mathrm{MGB}} > 0$) && ($T_{\mathrm{MGB}} < T$) && ($Q_{\mathrm{BT}} \geq Q_{\mathrm{BT\_min}}$). The torque split ratio (u) in electric driving mode is shown in (6).

$$u = 1 \quad (6)$$

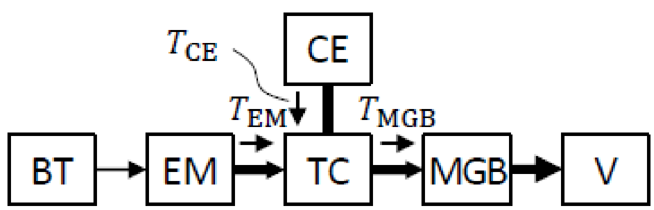

Fig. 6. Energy flow in Electric Driving mode [2]

E. Conventional Engine Mode

In this operational mode, the vehicle functions akin to a traditional vehicle, relying solely on the power generated by the engine to propel itself. This approach is utilized in situations where a moderate level of torque is necessary while maintaining a higher angular velocity. The torque split ratio (u) for conventional driving is depicted in (7).

$$u = 0 \tag{7}$$

V. ENERGY MANAGEMENT STRATEGY

Our methodology involves evaluating and comparing two separate control strategies for the Energy Management System (EMS) in a Parallel Hybrid Electric Vehicle (PHEV): a rule-based strategy and dynamic programming.

A. Rule Based Energy Management Strategy

The primary objective of Energy Management Systems is to achieve the optimal allocation of energy from various power sources based on the driver's requirements. The implementation of this objective is exemplified through the development of either global or local solutions [2], [7].

The rule-based strategy employs static controllers to perform fundamental control operations. These controllers are designed using if-else statements to account for different conditions or operating modes that must be met in order to achieve an optimal solution with minimized fuel consumption [7]. The integration of actual time supervisory control modes with cognitive abilities, heuristics, and simulation models occurs without taking into account prior knowledge of the driving cycle [7]. The optimal solution is derived from rule tables or flowcharts in order to fulfill the driver's specifications [7]. The operational mechanism of the rule-based strategy is elaborated in Fig. 7 [3].

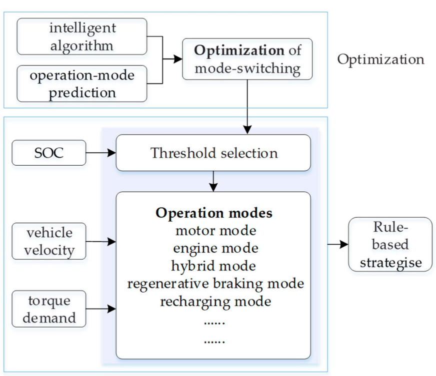

Fig. 7. Mechanism of Rule-Based Strategy [3]

This approach involves the activation of various operating modes based on specific inputs, including the battery's state of charge, the vehicle's velocity, and the driver's torque requirements [3]. In certain complex scenarios, the inclusion of system considerations, as demonstrated in the application of advanced algorithms in fuzzy rule-based strategies, can yield comprehensive solutions [3]. The proposed controller for the rule-based Strategy is shown in Fig. 8.

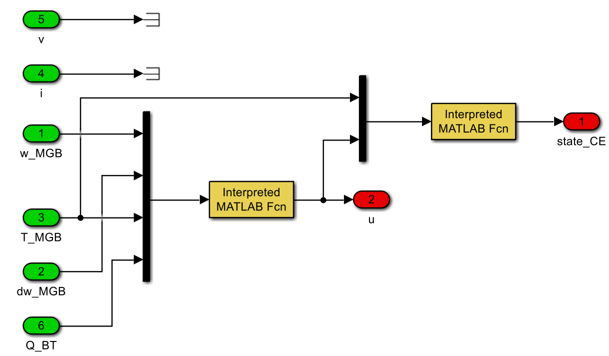

Fig. 8. Controller for Rule-Based Strategy

B. Dynamic Programming Approach

The previously mentioned vehicle operation modes significantly impact fuel consumption fluctuations during the drive cycle. The variable 'u' is exclusively responsible for determining the operational mode in effect. As previously demonstrated in the preceding section, the range of 'u' spans from -1 to 1. This study employs two driving cycles with respective cycle times of 1877 seconds and 1220 seconds [2]. To achieve an optimal outcome with regard to fuel consumption, it is necessary to have knowledge of the parameter 'u', which represents the operational mode in action, at each time step. The effectiveness of the variable 'u' in a rule-based strategy is compromised and can be restored by enhancing the degree of quantification. Moreover, this phenomenon leads to an escalation in the computational time and effort required to achieve the desired objective. The utilization of the Optimal Dynamic Programming (DP) strategy in this study is justified by its ability to achieve global optimality throughout the entire drive cycle, resulting in reduced fuel consumption.

The development of the Dynamic programming approach as a branch of Operations Research in 1951 was influenced by Bellman's principle [10]. Dynamic Programming is commonly employed in the analysis and resolution of complex problems that involve multiple stages of decision-making. It has proven to be an effective approach for finding optimal solutions to a wide range of practical problems across various domains [10].

Regardless of the intricacy of the problem, dynamic programming (DP) has the ability to offer an optimized solution [8]. In order to address the aforementioned assertion, the methodology employed in dynamic programming (DP) involves the segmentation of the problem process into a series of interrelated stages. This is subsequently followed by the identification of state variables, decision variables, and optimal value functions, with the aim of transforming the overarching problem into a collection of sub-problems that share the same characteristics [8]. The utilization of boundary conditions triggers a recursive calculation process, which has a tendency to yield an optimal solution [8]. The optimized outcomes derived from the preceding sub-problem are subsequently employed to determine a solution for the present sub-problem. Consequently, the solution obtained in the final sub-problem is considered the ultimate solution for the overall problem [8].

For this study, an optimization strategy was employed to determine the optimal torque distribution in a hybrid vehicle during the New European Driving Cycle (NEDC) and the Federal Test Procedure-75 (FTP-75). By discretizing state and control variables on a computational grid, the hybrid system's dynamics were captured using datasets representing engine, motor, and battery attributes. The DP method systematically computed the best control strategies, ensuring the battery state of charge remained within the desired limits while satisfying the driving demand for both cycles. The resulting optimized torque split ratios offer insights into the efficient operation of hybrid vehicles across diverse driving scenarios. Fig. 9 shows the proposed controller which makes use of optimized torque split vales to estimate the equivalent fuel consumption for both NEDC and FTP-75 driving cycle.

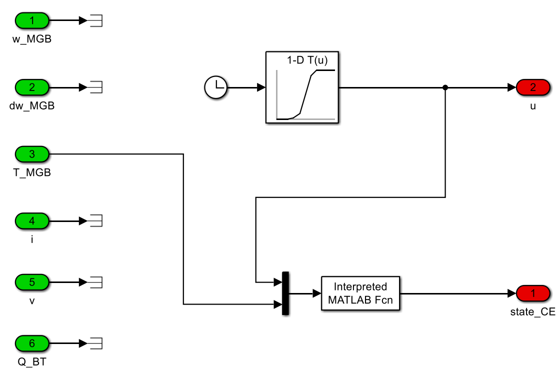

Fig. 9. Controller for Dynamic Programming Approach

Both strategies were implemented and tested separately to respect component constraints, such as maximum torques and currents. Their fuel efficiencies were compared with that of a conventional vehicle to gauge their effectiveness.

VI. RESULTS AND DISCUSSION

A. *Rule-Based Strategy*

The first part of the results section focuses on the Rule-Based Strategy. Table I delineates the outcomes of the Rule-Based Strategy when applied to two different driving cycles - the NEDC and FTP-75. Table I presents the Equivalent Fuel Consumption (in liters) and the State of Charge (in Coulombs) for each driving cycle.

TABLE I
RESULTS OF RULE-BASED ENERGY MANAGEMENT STRATEGY

Driving Cycle	Conventional Vehicle (Given)	Rule Based Strategy	
	Fuel Consumption (litres/100km)	Equivalent Fuel Consumption (litres/100km)	SOC (coulombs)
NEDC	4.897	3.559	18000
FTP-75	4.675	3.272	18005
Average (NEDC, FTP-75)	4.786	3.415	18002

Fig. 10 and Fig. 11 showcase the equivalent fuel consumption trends of the Parallel Hybrid Electric Vehicle (PHEV) while operating under the NEDC and FTP-75 driving cycles, respectively, utilizing a Rule-Based Strategy. These figures reveal a more elevated equivalent fuel consumption under the NEDC cycle when compared to the FTP-75 cycle.

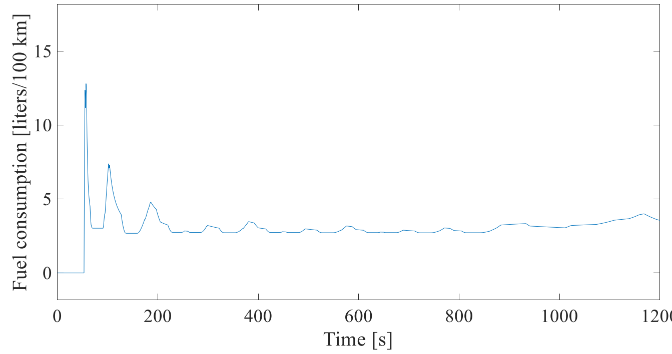

Fig. 10. Equivalent Fuel Consumption NEDC

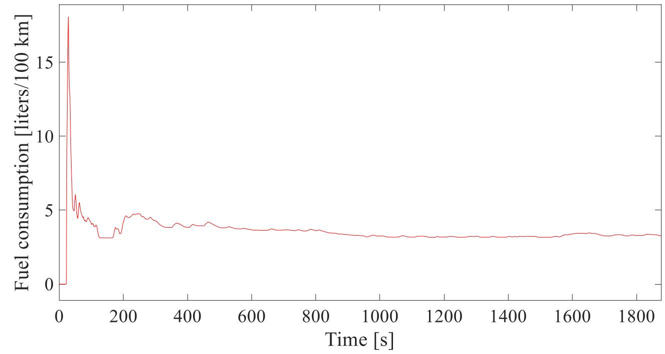

Fig. 11. Equivalent Fuel Consumption FTP-75

Fig. 12 and Fig. 13 trace the state of charge (SOC) patterns for the Parallel Hybrid Electric Vehicle (PHEV) under NEDC and FTP-75 driving cycles, respectively, following the Rule-Based Strategy. The graphs confirm a consistent initial and concluding SOC of 50 percent across both driving cycles.

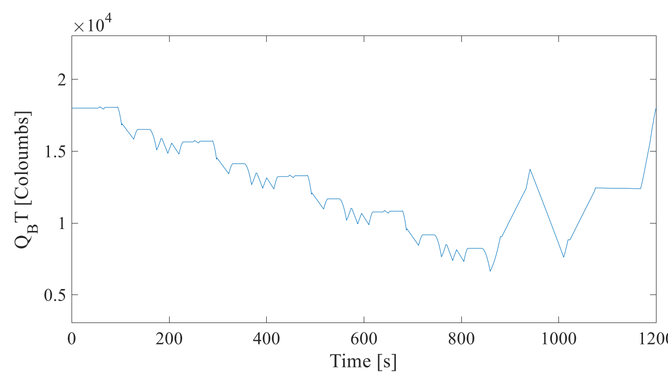

Fig. 12. State of Charge NEDC

Fig. 14 and Fig. 15 illustrate the fluctuations in the torque split ratio for a Parallel Hybrid Electric Vehicle (PHEV) under NEDC and FTP-75 driving cycles, respectively, as per the Rule-Based Strategy. The figures spotlight the changing

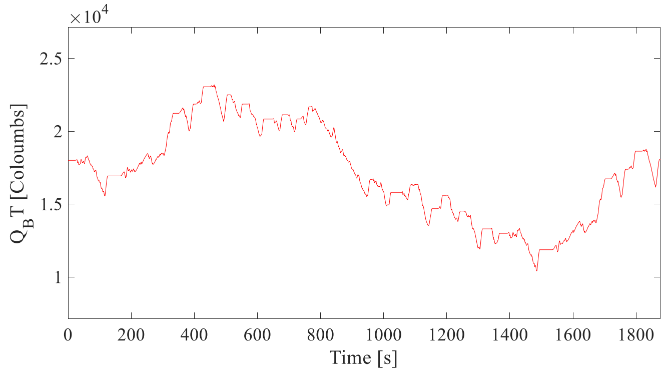

Fig. 13. State of Charge FTP-75

dynamics of the torque split ratio throughout various operating modes across both the NEDC and FTP-75 driving cycles.

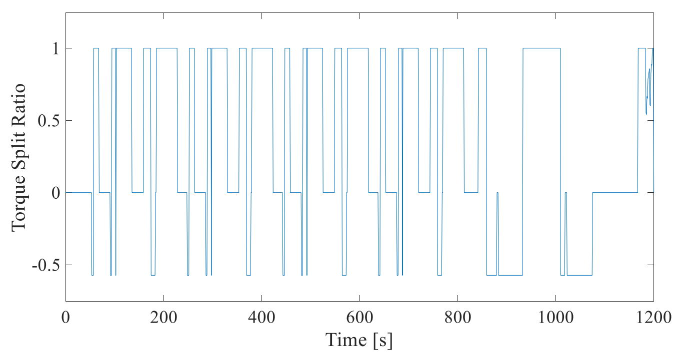

Fig. 14. Torque Split Ratio NEDC

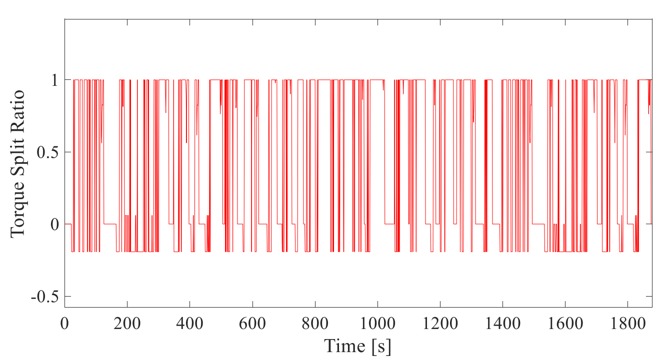

Fig. 15. Torque Split ratio FTP-75

B. Dynamic Programming Approach

The second subsection presents the results of the Dynamic Programming Approach, similarly applied to the NEDC and FTP-75 driving cycles. The Equivalent Fuel Consumption (litres) and State of Charge (Coulombs) for each driving cycle are outlined in Table II.

Fig. 16 and Fig. 17 graphically present the equivalent fuel consumption trajectories for the Parallel Hybrid Electric

TABLE II
RESULTS OF DYNAMIC PROGRAMMING APPROACH

Driving Cycle	Conventional Vehicle (Given)	Dynamic Programming	
	Fuel Consumption (litres/100km)	Equivalent Fuel Consumption (litres/100km)	SOC (coulombs)
NEDC	4.897	3.27	18001
FTP-75	4.675	2.998	18003
Average (NEDC, FTP-75)	4.786	3.134	18002

Vehicle (PHEV) under the NEDC and FTP-75 driving cycles, respectively, as computed using the Dynamic Programming approach. The illustrations suggest a pronouncedly higher equivalent fuel consumption under the NEDC driving cycle as opposed to the FTP-75 cycle.

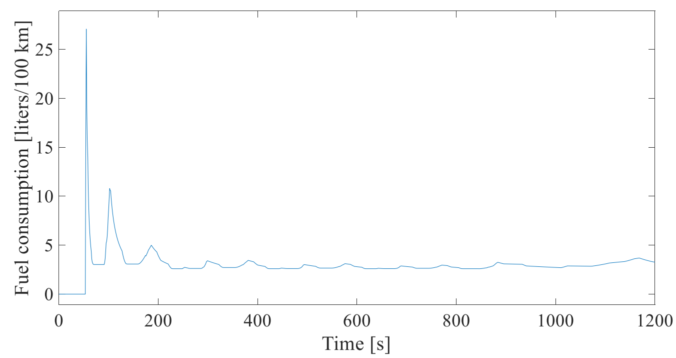

Fig. 16. Equivalent Fuel Consumption using Dynamic Programming-NEDC

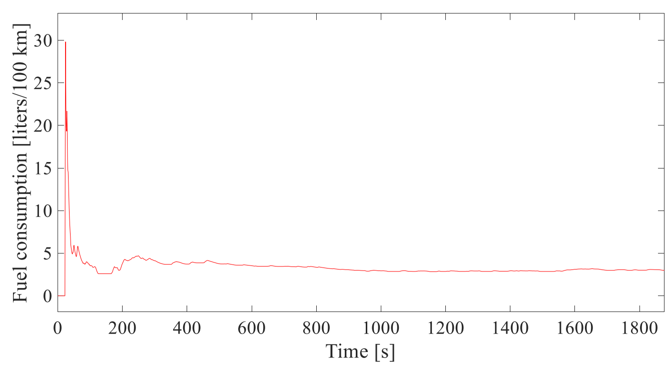

Fig. 17. Equivalent Fuel Consumption using Dynamic Programming-FTP-75

Fig. 18 and Fig. 19 depict the state of charge (SOC) evolution for the Parallel Hybrid Electric Vehicle (PHEV) during NEDC and FTP-75 driving cycles, respectively, as derived from the Dynamic Programming approach. These charts underscore a uniform commencement and termination SOC of 50 percent for both driving cycles.

Fig. 20 and Fig. 21 delineate the torque split ratio variations for a Parallel Hybrid Electric Vehicle (PHEV) navigating under the NEDC and FTP-75 driving cycles, respectively, as implemented through the Dynamic Programming approach.

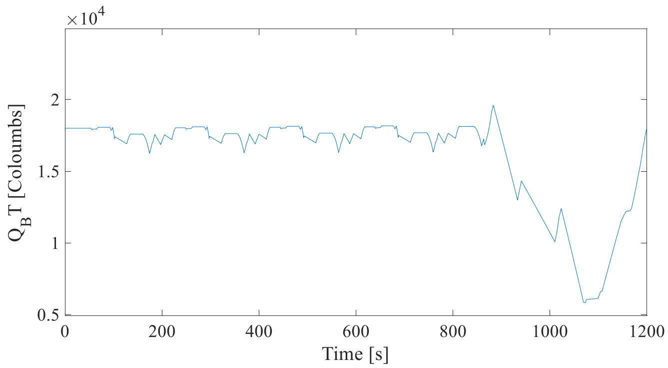

Fig. 18. State of Charge using Dynamic Programming-NEDC

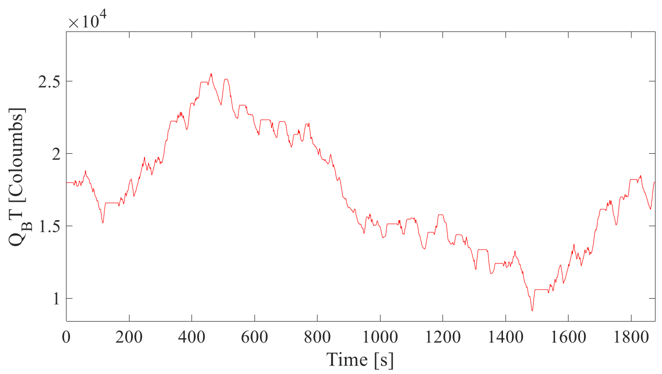

Fig. 19. State of Charge using Dynamic Programming-FTP-75

The figures convey the shifting landscape of the torque split ratio across distinct operational modes throughout both the NEDC and FTP-75 driving cycles.

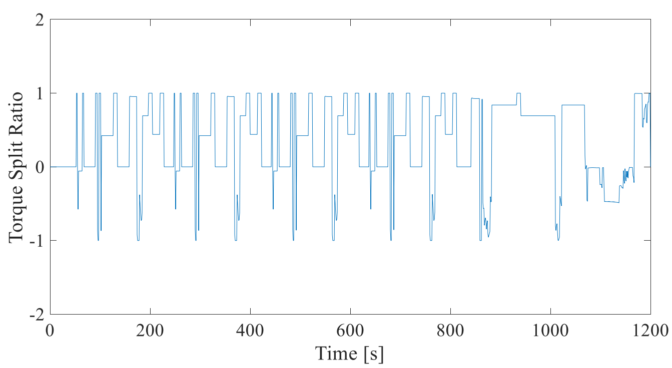

Fig. 20. Torque Split Ratio using Dynamic Programming-NEDC

C. Comparative Analysis between Rule-Based Strategy and Dynamic Programming

The third subsection provides a comparative analysis of the Rule-Based Strategy and Dynamic Programming Approach, as they are employed in the NEDC and FTP-75 driving cycles. The improvements in the efficiency of fuel consumption, as outlined in Table III, are calculated for both strategies in relation to the baseline conventional vehicle. These results

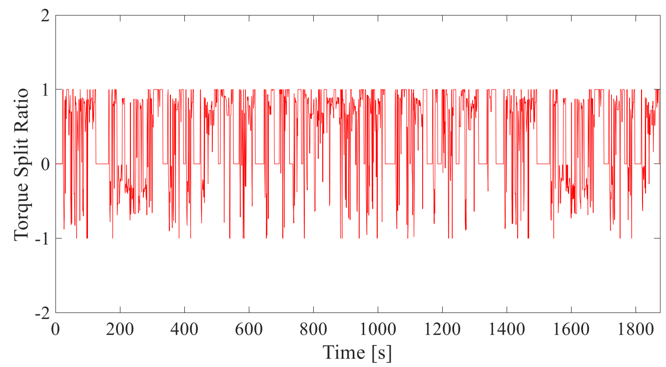

Fig. 21. Torque Split Ratio using Dynamic Programming-FTP-75

offer insights into the relative effectiveness of each approach in optimizing fuel consumption for the Parallel Hybrid Electric Vehicle (PHEV).

TABLE III
COMPARISON OF EFFICIENCY OF FUEL CONSUMPTION BETWEEN RULE-BASED STRATEGY AND DYNAMIC PROGRAMMING

Driving Cycle	Efficiency of Fuel Consumption Reduction (compared to conventional vehicle)	
	Rule-Based	Dynamic Programming
NEDC	27.32%	33.22%
FTP-75	30.01%	35.87%
Average (NEDC, FTP-75)	28.66%	34.54%

Fig. 22 illustrates a comparative analysis of fuel consumption trends in a Parallel Hybrid Electric Vehicle (PHEV), operating under the New European Driving Cycle (NEDC). The vehicle's performance is evaluated using two control strategies: Rule-Based Strategy and Dynamic Programming. Notably, the Dynamic Programming approach markedly minimizes fuel consumption relative to the Rule-Based Strategy, as the graph clearly substantiates.

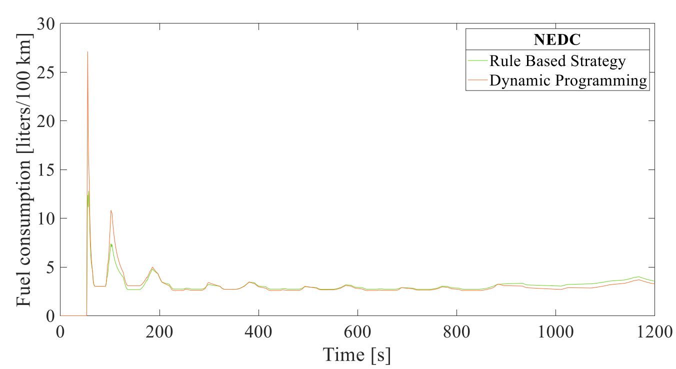

Fig. 22. Fuel Consumption Comparison for PHEV under NEDC: Rule-Based Strategy vs Dynamic Programming."

Fig. 23 presents a comparative analysis of the fuel consumption trends for a Parallel Hybrid Electric Vehicle (PHEV), while operating under the FTP-75 driving cycle. The performance of the vehicle is evaluated using two control strategies: Rule-Based Strategy and Dynamic Programming. It is clear from the graph that the Dynamic Programming method significantly reduces fuel consumption when compared to the Rule-Based Strategy.

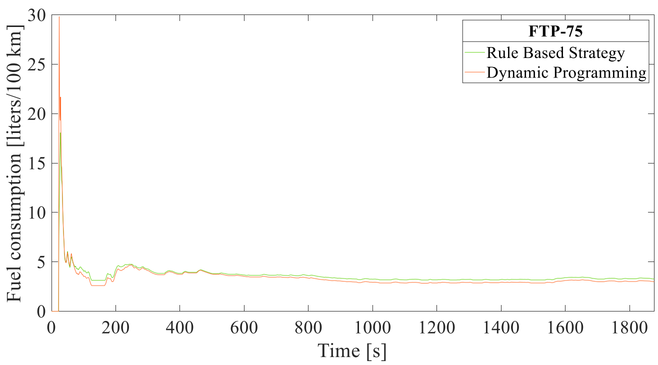

Fig. 23. Fuel Consumption Comparison for PHEV under FTP-75: Rule-Based Strategy vs Dynamic Programming

VII. Conclusion

In conclusion, this research provides an in-depth comparative study of the Rule-Based Strategy and the Dynamic Programming approach as energy management strategies for a Parallel Mild Hybrid Electric Vehicle (Mercedes-Benz A 170 CDI). Our findings demonstrate a substantial increase in fuel efficiency in both the NEDC and FTP-75 driving cycles, compared to a conventional vehicle. In particular, the Rule-Based Strategy resulted in an efficiency of fuel consumption reduction of $27.32\,\%$ and $30.01\,\%$ for the NEDC and FTP-75 cycles respectively. On average, the Rule-Based strategy shows $28.66\,\%$ efficiency for both the driving cycles. The Dynamic Programming approach, meanwhile, led to a remarkable fuel consumption reduction efficiency of $33.22\,\%$ and $35.87\,\%$ for the NEDC and FTP-75 cycles respectively. On average, the Dynamic Programming approach shows an efficiency of $34.54\,\%$for both the driving cycles which is about $6\,\%$ higher than the Rule-Based Strategy.

Furthermore, both strategies successfully maintained a $50\,\%$ state of charge at the beginning and end of each driving cycle, which indicates effective battery management. The torque split ratio demonstrated noticeable variations across different operational modes in both driving cycles. This underscores the potential of advanced energy management strategies in enhancing fuel efficiency for hybrid vehicles. Future research could focus on refining these strategies, incorporating more vehicle parameters like the gear ratio, and exploring other methods for further improvements in fuel consumption efficiency.

VIII. Future Scope

Addressing the identified limitations of both the Rule-Based Strategy and the Dynamic Programming approach will be pivotal for future research. The Rule-Based Strategy has shown promising results in enhancing fuel efficiency but has not taken into account the influence of gear ratios. Integrating the consideration of gear ratios in future studies could further refine the energy management strategy and uncover additional avenues for enhancing efficiency. Meanwhile, the Dynamic Programming approach, despite its remarkable efficiency improvements, relies on the availability of future information, limiting its practicality in real-world applications. Developing strategies that combine the benefits of Dynamic Programming without requiring future information will be essential. This might include the exploration of adaptive algorithms and machine learning models that can predict and adjust to real-time changes in driving conditions, making the strategies more versatile and widely applicable. Furthermore, investigating and incorporating a broader range of vehicle parameters and exploring alternative advanced energy management strategies will be vital. These advancements will contribute to the development of more robust and efficient energy management strategies for Parallel Mild Hybrid Electric Vehicles, pushing forward the frontiers of sustainable automotive technology.

References

[1] C. C. Chan and Y. S. Wong, ”The state of the art of electric vehicles technology,” 2004 The 4th International Power Electronics and Motion Control Conference, IPEMC, 2004, vol.1, pp. 46-57.

[2] D. Görges. (2021). Seminar Electromobility (summer term 2021) [pdf]. Available: https://olat.vcrp.de/auth/RepositoryEntry/3183214860/CourseNode/86162462107173.

[3] N. Xu et al., “Towards a Smarter Energy Management System for Hybrid Vehicles: A Comprehensive Review of Control Strategies,” Applied sciences, 2019, pp. 01-38.

[4] T. Hofmann, M. Steinbuch, R.M. van Druten and A. F. A Serrarens, “Rule- Based Energy Management Strategies for Hybrid Electric Drivetrains: A Fundamental Approach in Reducing Computation Time,” IFAC Proceedings Volumes, 2006, vol. 39, pp. 740-745.

[5] O. Govardhan, “Fundamentals and Classification of Hybrid Electric vehicles,” International Journal of Engineering and Techniques, 2017, vol. 3, pp. 194-198.

[6] O. Sundstrom and L. Guzzella, ”A generic dynamic programming Matlab function,” 2009 IEEE Control Applications, (CCA) Intelligent Control, (ISIC), 2009, pp. 1625-1630, doi: 10.1109/CCA.2009.5281131.

[7] V. A. Katkar and P. Goswami, “Review on Energy Management Systems for Hybrid E Vehicles,” 2020 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS), 2020, pp. 1-6. doi: 10.1109/ICPECTS49113.2020.9336977.

[8] L. Bauer, “Distance-based Optimization of 48V Mild-Hybrid Electric Vehicle,” M.S. thesis, Mechanical Engineering Dept., The Ohio State Univ., Athens, United States, 2018 [Online].

[9] Y. Tian, J. Liu,Q. Yao and K. Liu, “Optimal Control Strategy for Parallel Plug-in Hybrid Electric Vehicles Based on Dynamic Programming,” World Electric Vehicle Journal, 2021, pp. 0-17.

[10] T. G. Qin, W. Huang and G. X. Wang, “The application of Dynamic Programming in the Nonlinear Programming Problems,” 2010 WASE International Conference on Information Engineering, 2010, pp. 236-239. doi: 10.1109/ICIE.2010.151.